Process Management :—

① **Process** : The program under execution is called as process.
→ The program should reside in the main memory.
→ The program should occupy the CPU.

② **Attributes of Process** :

1. Process ID
2. Process State
3. Program Counter
4. Priority.
5. General Purpose Register
6. List of Open files.
7. List of Open devices.
8. Protection Information.
9. CPU Scheduling info.
10. Memory management information.
11. I/O Status info.
12. Accounting info.

① **Process id** : It is a unique identification no. which is assigned by the OS at the time of process creation.

② **Process State** : Contains the current state information of the process where it is reciding.

③ **Program Counter** : ~~Process contains the~~ The counter indicates the address of next instruction to be executed.

④ **Priority** : Parameter assigned by the OS at the time of process creation.

⑤ **List of Open files** : All the files opened by the process.

→ All these attributes of the process is called the content of a process.

→ Content of the process is stored in Process Controll Block (PCB).

PCB

Pid	P. State
PC	Priority
GPR	LOF
LOD	Protection info

→ Every process will have its own PCB.

→ PCB of the process will be stored in MM. (Main Memory).

→ Each process is represented in the operating system by Process Controll Block or task control block.

→ PCB contains many pieces of information related to a particular process.

* CPU Scheduling Information : This info includes the process priority, pointers to scheduling queues and scheduling parameters.

* Memory Management info : Value of the base and limit registers, page table, segment table etc.

* I/O Status information :- List of I/O devices allocated to the process, list of open files etc.

* Accounting information : Amount of the CPU and real time used, time limits, account numbers, job or process no etc.

⑥ Process State :

The process will have various states :

① New
② Ready
③ Running
④ Waiting OR Block
⑤ Terminated / Termination or Completion

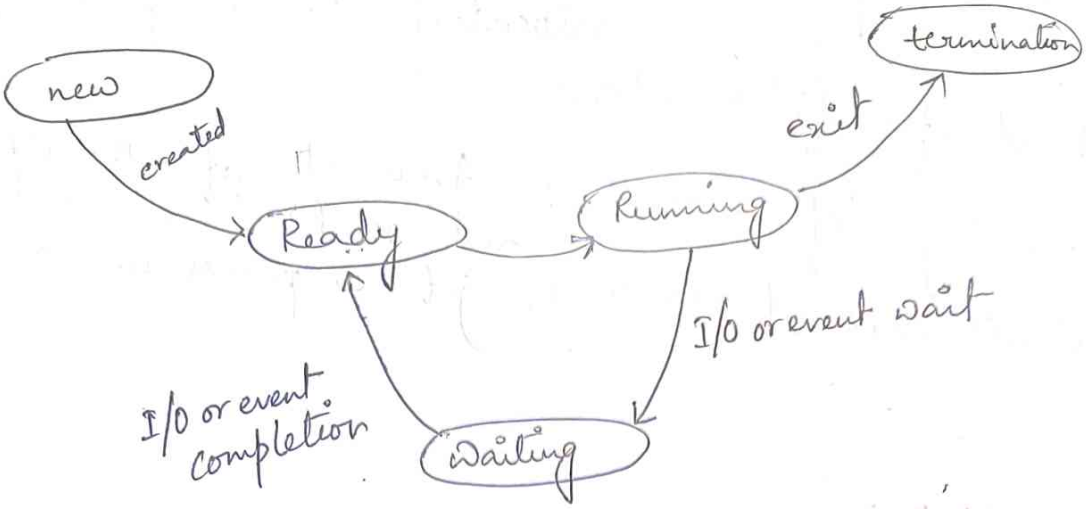
① New : The process is being created.

② Ready : The process is waiting to be assigned to CPU

③ Running : Instructions are being executed.

④ Wait : Process is waiting for some event to occure (such as I/O completion or reception)

⑤ Termination : The process is finished execution.



→ Initially the process will be in New state, i.e. the process is under creation or the process is created.

→ Once the process is created, it will move to Ready state. In Ready state there will be multiple no. of processes.

→ One of the processes will be selected from the Ready State and that will be scheduled or dispatched to Running state.

→ When the process is in Running State means it is executing in CPU. At running State, there will be only one process at any point of time.

→ When the Running process requires I/O operation, it will move to Wait state. The process will perform I/O time.

→ Once the I/O state is completed, the process will move to the Ready State.

→ When the running process completes execution, it will go to termination State.

Imp * Pre-emptive multiprogramming is multi-tasking.

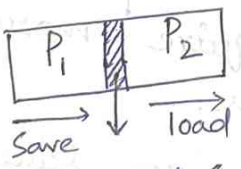
* In The pre-emptive OS, the high priority process is ~~sto~~ scheduled in running state and the low priority process is moved back to ready state from the running state.

* In non-preemptive OS, there is no priority, so the process can not return to Ready State from Running State.

* When the process is in Ready, Running, Wait State, it is residing in the main memory.

* Multiple Process ⟶ In Ready State and Wait State
  Only One Process ⟶ Running State.

Context Switching :—



Context Switching

→ Saving the context of one process and loading the context of another process is called as context switching.

Note: In special case, if only one process is there then also it is called as context switching
Ex. Round Robin Scheduling.

→ The context switching will also take some time.

→ The context switching time is considered as the overhead of the system.

⓪ Types of Process : (Wrt execution time)

- ① CPU Bound Process
- ② I/O Bound Process.

① CPU Bound Process :— The processes which require more time in CPU are called CPU Bound Process.

. The CPU Bound processes spends more time in the Running State.

② I/O Bound Process :— The processes which require more I/O time, are called I/O Bound process.

. I/O Bound process spends more time in Waiting State.

⓪ Degree of Multiprogramming :- The no. of processes present in the main memory at any point of time is called as degree of multiprogramming.

Schedulers in Operating System :—

→ The objective of multi-programing is to have some process running at all times, to maximum CPU utilization.

→ The objective of time-sharing is to swith the processes so frequently that users can interact with each program while it is running.

→ To meet these objectives, the process schedulers work. In OS, ~~select the available process~~ there are three types of schedulers :

1. Long-term Scheduler
2. Short-term Scheduler.
3. Mid-term Scheduler.

① Long-term Scheduler (LTS) OR Job Scheduler : LTS is responsible for creating and bringing the new process into the memory.

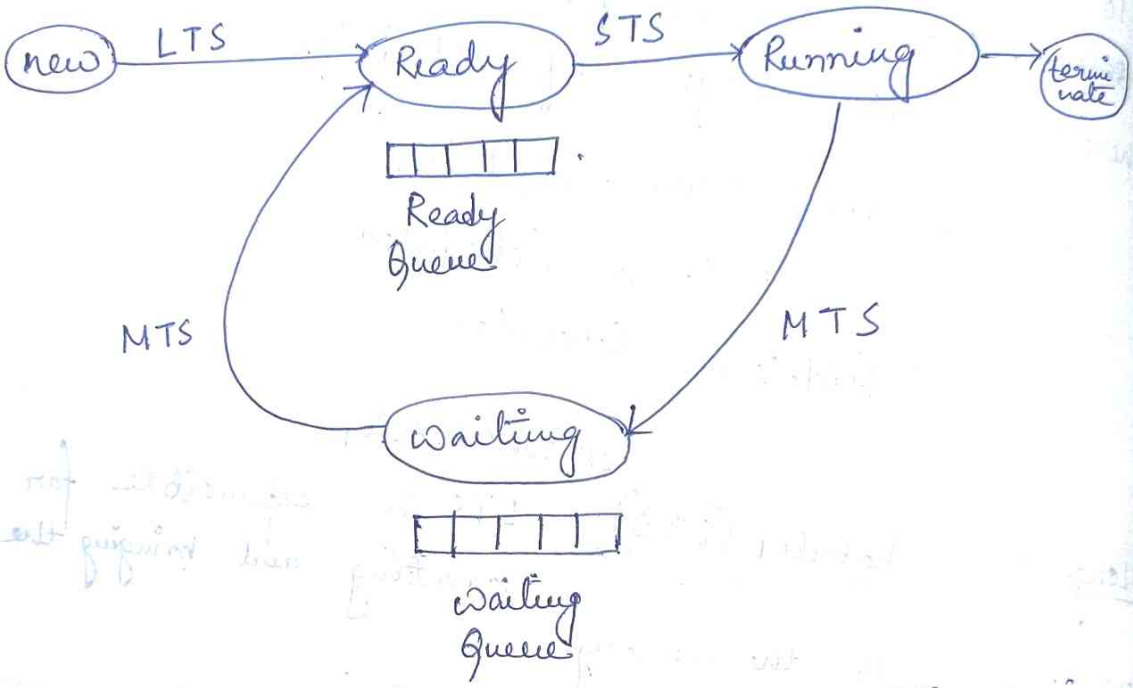
② Short term Scheduler (STS) OR CPU Scheduler :
Responsible for selecting one of the processes from ready state for scheduling on to Running state.

③ Mid-term Scheduler (MTS) :
Responsible for suspending and resuming the process.

→ Job done by MST is called as "Swaping".

• **Dispatcher** :— Dispatcher is responsible for saving the context of one process and loading the context of another process.

→ The context switching is done by the dispatcher.



* LST is responsible for degree of multiprograming.

Q Consider a system with 'n' CPU processors then what is the maximum no. of processes that may present in Ready, Running and Wait State?

Ans

State	Min	Max
Ready	0	All or any
Running	0	n
Wait	0	All or any

Process Scheduling :—

→ In a single-processor system, only one process can run at a time, any others must wait until the CPU is free and can be rescheduled.

→ The objective of multiprograming is to have some process running at all times, to maximize the CPU utilization.

CPU Scheduler :—

→ Whenever the CPU becomes idle, the operating System must select one of the processes in the ready queue to be executed.

→ The selection process is carried out by STS or CPU Scheduler.

→ The scheduler selects a process from the processes in memory that are ready to execute and allocates the CPU to the process.

Scheduling Criteria :—

In choosing which scheduling algorithm to use in a particular situation, we must consider the properties of various algorithm.

① Burst time(BT):— The time required by the process to complete execution, is called the Burst time (BT) of the process.

→ Time for which the process executes and remains in CPU.

② Completion Time (CP) :— At what time the process is completing its execution is called the ~~execution~~ completion time.

③ Turn Around time (TAT) :— (Execution time + Wait time)

The time difference between completion time and arrival time of a process.

$$\boxed{TAT = CT - AT}$$

→ The interval from the time of submission of a process and the time of completion.

④ Waiting time(WT):—

→ Waiting time is the sum of periods spent waiting in the ready queue.

→ The time difference between TAT and BT is called as WT.

$$\boxed{WT = TAT - BT}$$

⑤ Response Time (RT) :—

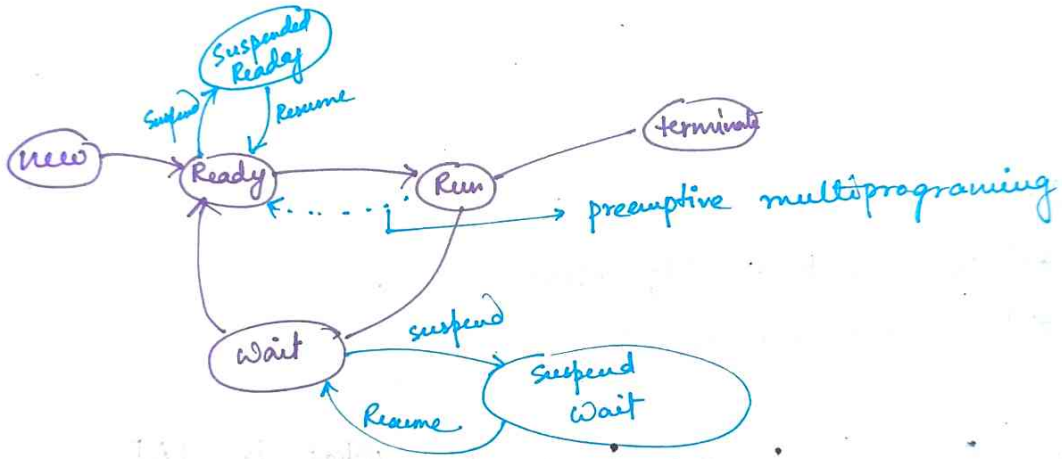
The time difference between the 1st response and the arrival time.

⑥ Throughput : No. of processes completed per unit time.

◉ Some New Concept of Process State Diagram :

① Suspended Ready

② Suspended Wait.



⇒ When the system resource is becoming short to manage process in the ready state, then some of the processes will be suspended and suspended processes will be moved to suspend Ready State.

⇒ When processes is in suspend Ready it will be residing in the backing store (secondary memory).

⇒ When the system resources becomes short to manage the process in wait state, then some of the processes will be moved to suspend wait state

⇒ When process is in the suspend wait, it will be residing in ~~blocking~~ ~~state~~ backing store (secondary mm)

PROBLEMS ON PROCESS SCHEDULING ALGORITHMS

FCFS

Problem 1

PNo (Pid)	AT	BT	TAT
1	0	4	4
2	1	3	6
3	2	1	6
4	3	2	7
5	4	5	11

What is TAT? WT?

Mode: Non-Preemptive

Gantt Chart

P_1	P_2	P_3	P_4	P_5

0 4 7 8 10 15

* Calculation of $\boxed{TAT = CT - AT}$

$P_1 = (4-0) = 4$

$P_2 = (7-1) = 6$

$P_3 = (8-2) = 6$

$P_4 = (10-3) = 7$

$P_5 = (15-4) = 11$.

$$\text{Avg TAT} = \frac{4+6+6+7+11}{5} = \frac{34}{5} = 6.8$$

* Calculate $\boxed{WT = TAT - BT}$

$P_1 = 4 - 4 = 0$

$P_2 = 6 - 3 = 3$

$P_3 = 6 - 1 = 5$

$P_4 = 7 - 2 = 5$

$P_5 = 11 - 5 = 6$

$$\therefore \text{Avg } WT = \frac{19}{5} = 3.8.$$

Q2

②

PNO.	AT	BT	TAT
1	6	4	
2	2	5	
3	3	3	
4	1	1	
5	4	2	
6	5	6	

CPU idle

	P_4	P_2	P_3	P_5	P_6	P_1

0 1 2 7 10 12 18 22

* Calculation $\boxed{TAT = CT - AT}$

$P_1 = (22 - 6) = 16$

$P_2 = (7 - 2) = 5$

$P_3 = (10 - 3) = 7$

$P_4 = (2 - 1) = 1$

$P_5 = (12 - 4) = 8$

$P_6 = (18 - 5) = 13$

$\therefore$ Avg TAT $= \dfrac{16 + 5 + 7 + 1 + 8 + 13}{6} = 8.33$

* Calculation $\boxed{WT = TAT - BT}$

$P_1 = (16 - 4) = 12$

$P_2 = (5 - 5) = 0$

$P_3 = (7 - 3) = 4$

$P_4 = (1 - 1) = 0$

$P_5 = (8 - 2) = 6$

$P_6 = (13 - 6) = 7$

$\therefore$ Avg WT $= \dfrac{12 + 0 + 4 + 0 + 6 + 7}{6} = 4.83$

ROUND ROBIN SCHEDULING (RR)

→ A concept of time quantum is considered.
→ Pre-emptive Scheduling
→ A concept of Ready Queue is considered. The ready Queue is considered as circular Queue.

Ex 1.

Pno.	AT	BT
1	0	4
2	1	5
3	2	2
4	3	1
5	4	6
6	6	3

TQ = 2

P_1	P_2	P_3	P_1	P_4	P_5	P_2	P_6	P_5	P_2	P_6	P_5

0 2 4 6 8 9 11 13 15 17 18 19 21

- Ready Queue

time 0 → ~~P1~~
2 → ~~P2~~ P3 P1
4 → ~~P3~~ P1 P4 P5 P2
6 → ~~P1~~ P4 P5 P2 P6 (P3 Finish)
8 → ~~P4~~ P_5 P_2 P_6 (P_1 Finish)
9 → ~~P5~~ P_2 P_6 (P_4 Finish)
11 → ~~P2~~ P_6 P_5
13 → ~~P6~~ P_5 P_2
15 → ~~P5~~ P_2 P_6

time 17 → ~~P2~~ P6 P5
18 → ~~P6~~ P_5 (P_2 Finish)
19 → ~~P5~~

- TAT ⇒

$P_1 = 8$

$P_2 = 17$

$P_3 = 4$

$P_4 = 6$

$P_5 = 17$

$P_6 = 13$

$\therefore$ Avg TAT = 10.83

- WT ⇒

$P_1 = 4$

$P_2 = 12$

$P_3 = 2$

$P_4 = 5$

$P_5 = 11$

$P_6 = 10$.

$\therefore$ Avg WT = 7.33

RR Scheduling

Ex 2 :

TQ = 3

Pno	AT	BT
1	5	~~5~~ ~~2~~ 0
2	4	~~6~~ ~~3~~ 0
3	3	~~7~~ ~~4~~ 1
4	1	~~9~~ ~~6~~ ~~3~~ 0
5	2	~~2~~ 0
6	6	~~3~~ 0

///	P_4	P_5	P_3	P_2	P_4	P_1	P_6	P_3	P_2	P_4	P_1	P_3

0 1 4 6 9 12 15 18 21 24 27 30 32 33

Ready Queue

0 → Idle

1 → ~~P_4~~

4 → ~~P_5~~ P_3 P_2 P_4

6 → ~~P_3~~ P_2 P_4 P_1 P_6 (P_5 Finish)

9 → ~~P_2~~ P_4 P_1 P_6 P_3

12 → ~~P_4~~ P_1 P_6 P_3 P_2

15 → ~~P_1~~ P_6 P_3 P_2 P_4

18 → ~~P_6~~ P_3 P_2 P_4 P_1

21 → ~~P_3~~ P_2 P_4 P_1 (P_6 Finish)

24 → ~~P_2~~ P_4 P_1 P_3

27 → ~~P_4~~ P_1 P_3 (P_2 Finish)

30 → ~~P_1~~ P_3 (P_4 Finish)

32 → ~~P_3~~ (P_1 Finish

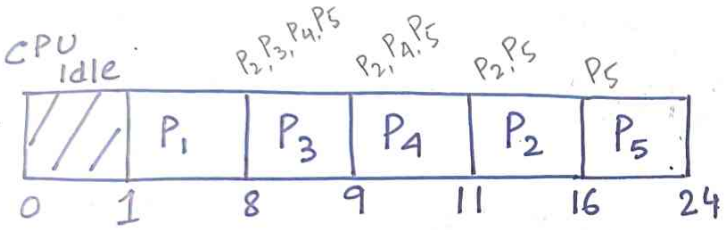
33 → (P_3 Finish)

SHORTEST JOB FIRST (SJF)

- Process Having Shortest BT will come first
- Non-Preemptive
- Whenever there is a conflict, Always follow FCFS.

(1)

Pid	AT	BT	TAT
1	1	7	7
2	2	5	14
3	3	1	6
4	4	2	7
5	5	8	19



* Calculation $\boxed{TAT = (CT - AT)}$

$P_1 = (8 - 1) = 7$

$P_2 = (16 - 2) = 14$

$P_3 = (9 - 3) = 6$

$P_4 = (11 - 4) = 7$

$P_5 = (24 - 5) = 19$

Avg TAT = 10.6

* Calculation $\boxed{WT = TAT - BT}$

$P_1 = (7) - 7 = 0$

$P_2 = (14 - 5) = 9$

$P_3 = (6 - 1) = 5$

$P_4 = (7 - 2) = 5$

$P_5 = (19 - 8) = 11$

$$\text{Avg } WT = \frac{9 + 5 + 5 + 11}{5} = 6$$

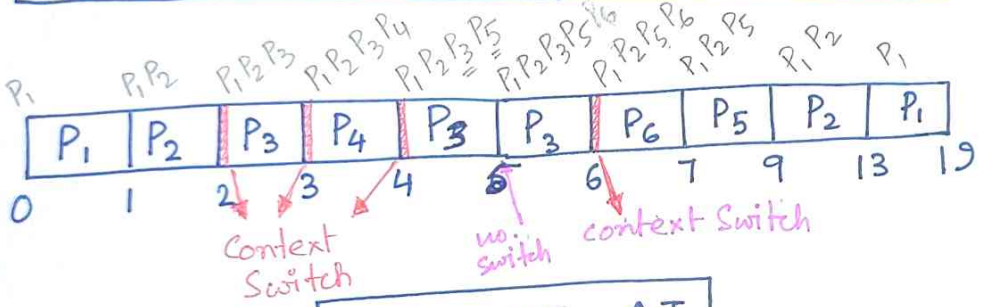
SRTF (SHORTEST REMAINING TIME FIRST)

→ Whenever the mode of SJF is <u>preemptive</u>, it becomes SRTF.

Q1

PNo.	AT	BT	CT	TAT	WT
1	0	~~7~~ ~~6~~ 0	19	19	12
2	1	~~5~~ ~~4~~ 0	13	12	7
3	2	~~3~~ ~~2~~ ~~1~~ 0	6	4	1
4	3	~~1~~ 0	4	1	0
5	4	~~2~~ 0	9	5	3
6	5	~~1~~ 0	7	2	1

* Let, every process executes for One time Unit.
* In Preemptive SJF, Process get ~~preemptive~~ <u>preempted</u> each time, a new process arrives.



* At 4, P_3 and P_5 both have same BT. So, follow FCFS

* Calculate $\boxed{TAT = CT - AT}$

$P_1 = (19 - 0) = 19$

$P_2 = (13 - 1) = 12$

$P_3 = (6 - 2) = 4$

$P_4 = (4 - 3) = 1$

$P_5 = (9 - 4) = 5$

$P_6 (7 - 5) = 2$

Avg TAT = 7.16 (Ans).

* * * Here, total no. of <u>context switch = 8.</u>

Any time CPU changes from one process to other, context switch. — whether the change due to preemption or process completion

* Calculate $\boxed{WT = TAT - BT}$

$P_1 = (19 - 7) = 12$

$P_2 = (12 - 5) = 7$

$P_3 = (4 - 3) = 1$

$P_4 = (1 - 1) = 0$

$P_5 = (5 - 2) = 3$

$P_6 = (2 - 1) = 1$

∴ Avg WT = 4 (Ans)

Example of SJF and SRTF

Ex: SJF

PNO.	AT	BT
1	6	1
2	3	3
3	4	6
4	1	5
5	2	2
6	5	1

~~TAT → TAT = CT − AT~~ ③

~~$P_1 = 7-6 = 1$~~

~~$P_2 = (13-3) = 10$~~

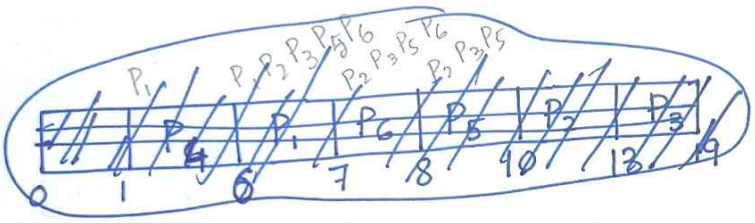
~~$P_3 = (19-4) = 15$~~

~~$P_4 = (6-1) = 5$~~

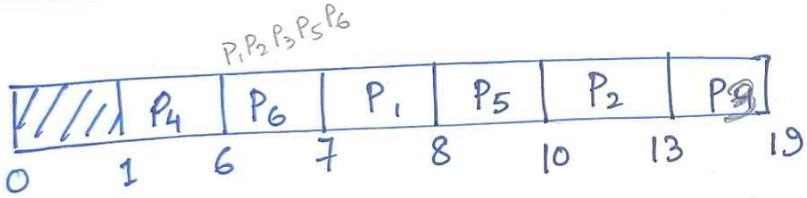
~~$P_5 = (10-2) = 8$~~

~~$P_6 = 8-5 = 3$~~

~~Avg TAT = 7~~



$\boxed{WT = TAT - BT}$



TAT:

$P_1 = (8-6) = 2$

$P_2 = (13-3) = 10$

$P_3 = (19-4) = 15$

$P_4 = (6-1) = 5$

$P_5 = (10-2) = 8$

$P_6 = (7-5) = 2$

$$\text{Avg TAT} = \frac{2+10+15+5+8+2}{6} = 7 \text{ (Ans)}$$

WT: $\boxed{WT = TAT - BT}$

$P_1 = (2-1) = 1$

$P_2 = (10-3) = 7$

$P_3 = (15-6) = 9$

$P_4 = (5-5) = 0$

$P_5 = (8-2) = 6$

$P_6 = (2-1) = 1$

$$\therefore \text{Avg WT} = \frac{0+1+1+6+7+9}{6} = \frac{24}{6} = 4 \text{ (Ans)}$$

Ex. SRTF

(4)

One Unit time time Share

PNO	AT	BT
1	3	~~4~~
2	4	~~2~~ 0
3	5	~~1~~ 0
4	2	~~6~~ 0
5	1	~~8~~ ~~7~~ 0
6	2	~~4~~ ~~3~~ ~~2~~ ~~1~~

P_5 $P_5 P_4 P_6$ $P_5 P_4 P_6 P_1$ $P_5 P_4 P_6 P_1 P_2$ (FCFS) $P_5 P_4 P_6 P_1 P_2 P_3$ (FCFS) $P_5 P_4 P_1 P_2 P_3$ $P_5 P_4 P_1 P_2$ $P_5 P_4 P_1$ $P_5 P_4$

	P_5	P_6	P_6	P_6	P_6	P_3	P_2	P_1	P_4	P_5

0 1 2 3 4 5 6 7 9 13 19 26

- **TAT**

$P_1 = (13-3) = 10$

$P_2 = (9-4) = 5$

$P_3 = (7-5) = 2$

$P_4 = (19-2) = 17$

$P_5 = (26-1) = 25$

$P_6 = (6-2) = 4$

Avg TAT = 10.5

- **WT**

$P_1 = (10-4) = 6$

$P_2 = (5-2) = 3$

$P_3 = (2-1) = 1$

$P_4 = (17-6) = 11$

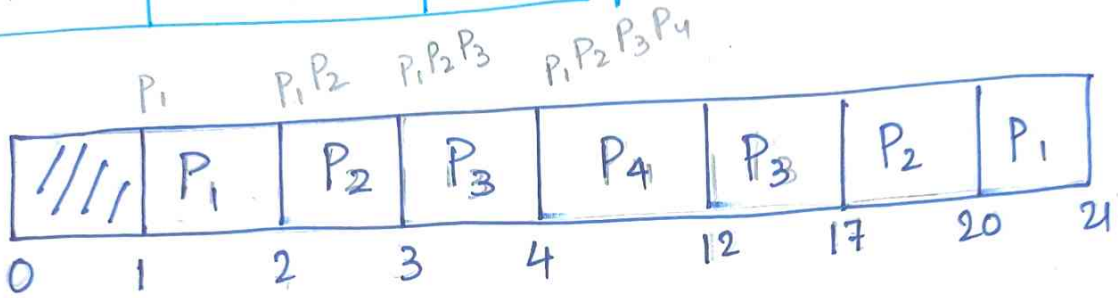
$P_5 = (25-8) = 17$

$P_6 = (4-4) = 0$

Avg WT = 6.33

LRTF (Longest Remaining Time First)

Process	AT	BT
P1	1	~~2~~ 1
P2	2	~~4~~ 3
P3	3	~~6~~ 5
P4	4	~~8~~



TAT

$P_1 = 21 - 1 = 20$

$P_2 = 20 - 2 = 18$

$P_3 = 17 - 3 = 14$

$P_4 = 12 - 4 = 8$

$\therefore$ Avg TAT $= 15$

WT

$P_1 = 20 - 2 = 18$

$P_2 = 18 - 4 = 14$

$P_3 = 14 - 6 = 8$

$P_4 = 8 - 8 = 0$

$\therefore$ Avg WT $= 10$.

LONGEST JOB FIRST (LJF)

①

* Non-Preemptive

Pno	AT	BT
1	0	3
2	1	2
3	2	4
4	3	5
5	4	6

$\boxed{TAT = CT - AT}$

$P_1 = 3 - 0 = 3$

$P_2 = 20 - 1 = 19$

$P_3 = 18 - 2 = 16$

$P_4 = 8 - 3 = 5$

$P_5 = 14 - 4 = 10$

Avg TAT = 10.6

P_1	P_4	P_5	P_3	P_2

0 3 8 14 18 20

$\boxed{WT = TAT - BT}$

$P_1 = 3 - 3 = 0$

$P_2 = 19 - 2 = 17$

$P_3 = 16 - 4 = 12$

$P_4 = 5 - 5 = 0$

$P_5 = 10 - 6 = 4$

$$\text{Avg. } WT = \frac{0 + 17 + 12 + 0 + 4}{5} = 6.6$$

Preemptive Priority Scheduling

	Priority	Pno	AT	BT
Low →	4	1	1	~~4~~ 3
	5	2	2	2
	7	3	2	~~3~~ 2
High →	8	4	3	~~5~~ ~~4~~ 0
	5	5	3	1
	6	6	4	2

$P_1 P_2 P_3$ $P_1 P_2 P_3 P_4 P_5$ $P_1 P_2 P_3 P_4 P_5 P_6$ $P_1 P_2 P_3 P_5 P_6$

////	P_1	P_3	P_4	P_4	P_3	P_6	P_2	P_5	P_1

0 1 2 3 4 8 10 12 14 15 18

Avg TAT = 10.3

Avg WT = 7.5

PRIORITY SCHEDULING

→ Criteria : Priority value

→ Mode : Can have Pre-emptive and Non-preemptive both versions.

Non-Preemptive Priority Scheduling

Ex

	Priority	Pro.	AT	BT
	4	1	0	4
	5	2	1	5
Highest →	7	3	2	1
	2	4	3	2
Lowest →	1	5	4	3
	6	6	5	6

$P_2 P_3 P_4 P_5$ $P_2 P_4 P_5 P_6$ $P_2 P_4 P_5$ $P_4 P_5$

P_1	P_3	P_6	P_2	P_4	P_5

0 4 5 11 16 18 21

• TAT

$P_1 = 4 - 0 = 4$

$P_2 = 16 - 1 = 15$

$P_3 = 5 - 2 = 3$

$P_4 = 18 - 3 = 15$

$P_5 = 21 - 4 = 17$

$P_6 = 11 - 5 = 6$.

Avg TAT = 10. (Ans)

$\therefore$ Avg WT $= 6.5 = \frac{39}{6} = 6.5$ (Ans)

4 HIGHEST RESPONSE RATIO NEXT (HRRN)

①

→ Highest Response Ratio Next (HRRN) is a non-preemptive CPU Scheduling in OS.

→ Aims to improve fairness and avoid starvation while still giving preference to shorter jobs.

→ Balances between SJF and FCFS.

→ In this process, next process scheduled for execution is one with the Highest Response Ratio among all ready (waiting) processes.

$$\boxed{\text{Response Ratio (RR)} = \frac{w+s}{s} = 1 + \frac{w}{s}}$$

where,

w = waiting time of the process so far. (WT)

s = Estimated Service time (BT).

→ It is a non-preemptive process.

Ex. HRRN

Process	AT	BT	TAT	WT
P_1	0	2	2	0
P_2	2	6	6	0
P_3	4	7	6	3
P_4	5	3	14	7
P_5	7	5	16	11

P_1	P_2	$P_3 P_4 P_5$	$P_3 P_5$	
P_1	P_2	P_4	P_3	P_5

0 2 8 11 18 23

- **At time = 8**

$WT\ P_3 = 8 - 4 = 4$

$$\boxed{RR = \left(\frac{WT + BT}{BT}\right)}$$

$WT\ P_3 = \frac{4+7}{7} = 1.57$

$WT\ P_4 = 8 - 5 = 3$

$RR = \frac{3+3}{3} = 2$

$WT\ P_5 = (8-7) = 1$

$RR = \frac{1+5}{5} = \frac{6}{5} = 1.2$

So, P_4 has highest RR.

- **At time = 11**

$WT\ P_3 = 11 - 4 = 7$

$RR = \frac{7+7}{7} = 2$ (Highest RR)

$WT\ P_5 = (11-7) = 4$

$RR = \frac{4+5}{5} = \frac{9}{7} = 1.8$

$\therefore$ Avg TAT $= \frac{2+6+6+14+16}{5} = 8.8$

$\therefore$ Avg WT $= \frac{0+0+3+7+11}{5} = 4.2$

RN

Ex 2

Process	AT	BT
P1	0	8
P2	1	4
P3	2	2
P4	3	1

$P_2 P_3 P_4$

P_1	P_4	P_3	P_2

0 8 9 11 15

At 8

$RR(P_2) = 1 + \frac{7}{4} = \frac{11}{4} = 2.75$

$RR(P_3) = 1 + \frac{6}{2} = \frac{8}{2} = 4$

$RR(P_4) = 1 + \frac{5}{1} = \frac{8}{1} = 6$

At 9

$RR(P_2) = 1 + \frac{8}{4} = 3$

$RR(P_3) = 1 + \frac{7}{2} = 4.5$

$\therefore$ Avg TAT = 9.25

Avg WT = 5.5

MULTILEVEL QUEUE SCHEDULING

→ Here, ready queue is divided into multiple Separate Queues, each for different class or type of processes (like system, interactive, batch etc).

→ Each queue can have its own scheduling algorithm and has a priority level.

→ Process are assigned to a queue based on characteristics (eg. priority, type, memory usage).

→ Once assigned the processes are remains inside that queue (No moving between queues in basic MLQ)

→ Each queue can use different scheduling algorithm (eg. RR, FCFS, SJF).

→ Queues themselves have priority, higher priority queues are considered before lower ones.

→ <u>Two Main Steps</u> :

1. Assign Process to Queue : Based on process type (ex. system, interactive, batch etc.)

2. Schedule Queues and Processes :
   - Scheduler picks the highest Priority non-empty queue.
   - Within that queue, it uses queue's own scheduling algorithm.

Ex : Suppose there are three Queues.

Queue Level	Process Type	Scheduling Algo
Q_1	System Process	Round Robin (RR)
Q_2	Interactive Process	SJF
Q_3	Batch Process	FCFS

Now imagine these processes :

Process	Queue	AT	BT
P_1	Q_1	0	4
P_2	Q_2	1	3
P_3	Q_1	2	2
P_4	Q_3	3	5

* Scheduling flow :

i] At time 0, only P_1 (Q_1) exists → it runs first using RR.

ii] At time 1, P_2 (Q_2) arrives, but Q_1 still has higher priority

iii] At time 2, P_3 (Q_1) arrives, → stays in higher priority Queue

iv] Q_1 Processes P_1 and P_3 before switching to Q_2. Only when Q_1 empty. Q_2 starts.

v] After Q_1, P_2 runs then finally P_4 from Q_3 executes once only Q_1 and Q_2 are empty.

So, the CPU always prefers higher-priority Queue first.

MLQ

Process	AT	BT	Queue	Scheduling Algo
P1	0	5	Q_1	FCFS (Highest Priority)
P2	1	8	Q_2	RR (Time Quantum = 2
P3	2	4	Q_3	SJF
P4	3	6	Q_2	RR (time quantum = 2

Queue Priority :

$Q_1 > Q_2 > Q_3$

P_1	P_2 (P_2, P_4)	P_4	P_2	P_4	P_2	P_4	P_2	P_3

0 5 7 9 11 13 15 17 19 23

- Time 0-5 : Q_1 (Highest Priority - FCFS)
  Only P_1 at time 0, → executes first.
- Time 5 onwards : Q_2 (RR, TQ = 2).

* After P_1 finishes, higher priority queue Q_1 is empty. So go to next priority Q_2. Q_2 has P_2 and P_4.

* RR scheduling in Q_2 (TQ = 2)
At time 19, Q_2 finishes (As P_2 and P_4 finishes)

- Time 19 onwards : Q_3 (SJF)
  At 23, Q_3 finishes. (As P_3 finishes).

⑤ Multi-level Queue Scheduling :—

→ Applyed where processes are easily classified into different groups. Ex: foreground (interactive) and background (batch) processes.

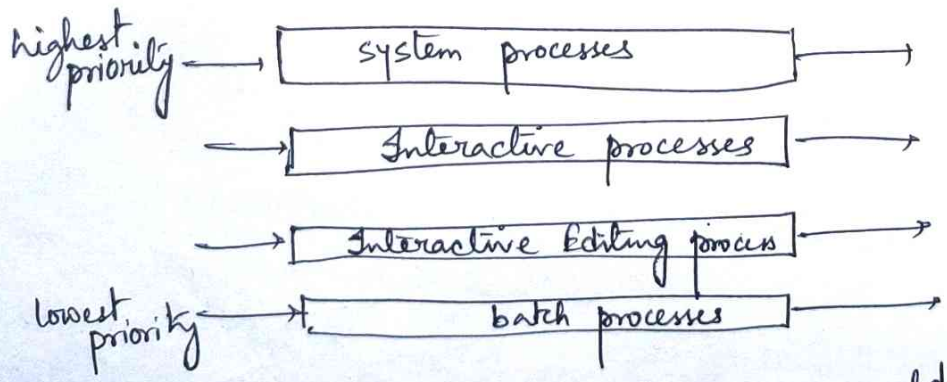
→ These two types of processes have different response time requirements and so, may have different scheduling needs. Foreground processes may have priority over background processes.

→ A multi-level queue scheduling partitions the ready queue in several ready queues.

→ The processes are permanently stored in one queue based on their properties. (memory requirements, process priority, process type).

→ Seperate queue has their own scheduling algorithm.

→ Additionally, there must be scheduling among queues, commonly used as fixed-priority preemptive scheduling. Ex. foreground process may have priority more than background.



→ No process in the batch queue could run untill All the previous queues are empty.

→ high priority processes will be placed in the top ready queue and the low priority processes will be placed in the bottom level. Ready Queue.

→ If this is the stategy followed, then the processes which are placed in the bottom level RQ will suffer from starvation.

* **Starvation** : The indefinite waiting of a process is called starvation.

⑥ Multi-level feedback Queue Scheduling :—

→ In multi-level queue scheduling, processes are assigned to each queue permanently.

→ In multi-level queue scheduling, processes do not move to one queue to other as the processes do not change their nature (say, foreground, background).

→ In multi-level feedback queue scheduling, allows a process to move between queues.

→ The idea is to separate processes according to the characteristics of their CPU bursts.

→ If a process uses too much of CPU time, it will be moved to lower priority queue.

→ This scheme leaves I/O bound and interactive processes to the higher priority queues.

→ A process that waits too long in a lower-priority queue, may be moved to a higher priority queue. This prevents <u>starvation</u>.